Thomas Andrew Lamb

PhD Student in Machine Learning, University of Oxford

thomas.lamb@eng.ox.ac.uk | https://tomalamb.github.io/

LinkedIn | GitHub | Google Scholar

Experience

Oxford, UK

- **Apple Machine Learning Research [🌐]** *May 2026 – Oct 2026*
  Research Scientist Intern Paris, France
  - Working with Sinead Williamson and Michael Kirchhof on post-training for budget-aware agents.
  - Internship culminated in the paper "Rewards for Budget-Aware Multi-Turn Agents That Adapt to Arbitrary Test-Time Constraints".

- **DeepMind / Torr Vision Group, University of Oxford [🌐]** *Jul 2022 – Jan 2023*
  Machine Learning Research Intern Oxford, UK
  - Designed a framework for faithful knowledge distillation.
  - Derived LP-based bounds on teacher–student confidence disagreement enabling formal analysis.
  - Developed methods yielding more calibrated student models; resulted in our paper *Faithful Knowledge Distillation*.

- **University of Durham [🌐]** *Jul 2019 – Sept 2019*
  Summer Research Student Durham, UK
  - Proved new identities relating polylogarithmic integrals to multiple zeta values with Prof. Herbert Gangl.

Education

- **University of Oxford** *Oct 2023 – Present*
  PhD (DPhil) in Machine Learning, Department of Engineering Science Oxford, UK
  - Supervisors: Tim G. J. Rudner (University of Toronto & Vector Institute) and Philip H. S. Torr (University of Oxford)
  - Research on AI safety and reliable ML, developing methods for LLM uncertainty, calibration, and generalisation.

- **University of Edinburgh** *Sept 2022 – Aug 2023*
  MSc Artificial Intelligence (Distinction) Edinburgh, UK
  - MSc Artificial Intelligence Class Prize (ranked 1st overall)
  - Dissertation: *Self-Supervised Learning of Tractable Generative Models*

- **University of Durham** *Oct 2016 – Jul 2020*
  MMath Mathematics (First-Class Honours) Durham, UK
  - Percy Heywood Prize; John Crowther Prize
  - Dissertation: *An Introduction to Modular Forms and the Eichler–Shimura Isomorphism*

Publications

Conference Papers

- **Tight PAC-Bayes Generalisation Guarantees for Large Language Model Safety Monitoring** *2026*
  T. A. Lamb, P. Torr, T. Rudner NeurIPS
- **Improving Semantic Uncertainty Quantification in Language Model Question-Answering via Token-Level Temperature Scaling** *2026*
  T. A. Lamb, D. Ivanova, P. Torr, T. Rudner AISTATS
- **Focus On This, Not That! Steering LLMs with Adaptive Feature Specification** *2025*
  T. A. Lamb, A. Davies, A. Paren, P. Torr, F. Pinto ICML
- **Universal In-Context Approximation by Prompting Fully Recurrent Models** *2024*
  A. Petrov, T. A. Lamb, A. Paren, P. Torr, A. Bibi NeurIPS
- **Hidden in Plain Sight: Evaluating Abstract Shape Recognition in Vision-Language Models** *2024*
  A. Hemmat, A. Davies, T. A. Lamb, J. Yuan, P. Torr, A. Khakzar, F. Pinto NeurIPS

- **Detecting LLM Hallucination through Layer-wise Information Deficiency: Analysis of Unanswerable Questions and Ambiguous Prompts** *2025*
  H. Kim, T. A. Lamb, A. Bibi, P. Torr, Y. Gal EMNLP

Workshops and Preprints

- **Rewards for Budget-Aware Multi-Turn Agents That Adapt to Arbitrary Test-Time Constraints** *2026*
  T. A. Lamb, A. Golinski, E. Gualdoni, S. Williamson, M. Kirchhof Preprint
- **Towards Label-Free Biological Reasoning Synthetic Dataset Creation via Uncertainty Filtering** *2025*
  J. L. Stoisser, L. Phillips, A. Misra, T. A. Lamb, P. Torr, M. B. Martell, J. Fauqueur, K. Martens NeurIPS ER
- **Can Large Language Models Express Uncertainty Like Humans?** *2025*
  L. Tao, Y. F. Yeh, B. Kai, M. Dong, T. Huang, T. A. Lamb, J. Yu, P. Torr, C. Xu arXiv
- **Faithful Knowledge Distillation** *2023*
  T. A. Lamb, R. Bunel, K. Dvijotham, M. P. Kumar, P. H. S. Torr, F. Eiras Preprint

Awards and Prizes

- **MSc Artificial Intelligence Class Prize** *2023*
  University of Edinburgh
  ◦ Awarded to the student who attained the highest overall mark in the MSc Artificial Intelligence course.
- **Percy Heywood Prize** *2020*
  Department of Mathematical Sciences, University of Durham
  ◦ Awarded to a student graduating with an MMath whose performance is outstanding in the final year.
- **John Crowther Prize** *2019*
  University College, University of Durham
  ◦ Awarded for performance in third-year Mathematics examinations.

Talks

- **Improving Semantic Uncertainty Quantification in LLMs via Temperature Scaling** *Sept 2025*
  Oxford Novo Nordisk
- **Semantic-Level Confidence Calibration of Language Models via Temperature Scaling** *Mar 2025*
  Oxford University of Oxford
- **Towards Safety and Transparency: Addressing Bias and Uncertainty Quantification in LLMs** *Oct 2024*
  London Huawei

Teaching

- **University of Oxford and Online** *Sept 2020 – Present*
  Mathematics and AI Tutor Oxford, UK & Remote
  ◦ Teach MSc-level Reinforcement Learning, Statistical Learning Theory (OPUS program) and tutor GCSE/A-Level and undergraduate mathematics.
- **University of Durham** *Oct 2019 – Jun 2020*
  Mathematics Teaching Assistant Durham, UK
  ◦ Marked and provided detailed feedback on second-year Algebra coursework, ensuring clarity and rigor.

References

- **Dr. Tim G. J. Rudner**
  Assistant Professor University of Toronto / Vector Institute
  ◦ Contact details: available upon request
- **Prof. Philip H. S. Torr**
  Professor of Engineering Science University of Oxford
  ◦ Contact details: available upon request